

isoSPI, UART Telemetry, and Fault Logic Design

Battery Management System for Electric Scooter | ECE 445 Team 35 | Technical Design Note

Communication Architecture

The BMS uses two communication layers. SPI is used locally between the STM32 and LTC6820, while isoSPI is used between the master board and the LTC6811 slave board. UART is used externally to stream processed telemetry from the master board to the BMS Viewer.

Link	Endpoints	Purpose	Reason for Selection
SPI	STM32 to LTC6820	Local command and data interface.	Fast, synchronous, simple board-level interface.
isoSPI	LTC6820 to LTC6811	Battery-monitor communication across boards.	Differential and transformer-isolated, improving noise immunity and breaking ground loops.
UART	STM32 to BMS Viewer	Telemetry and diagnostic data output.	Simple serial interface for PC-based visualization.

isoSPI Design Rationale

Battery packs are electrically noisy because of switching regulators, changing load current, balancing transients, and high di/dt paths. isoSPI is well suited to this environment because the signal is differential and transformer coupled.

- Differential signaling improves common-mode noise rejection.
- Transformer isolation improves safety and reduces ground-loop issues.
- The LTC6811 natively supports isoSPI, reducing protocol translation complexity.
- The design verified 1 MHz SPI activity and successful isoSPI communication to the LTC6811.

Telemetry Packet Structure

The firmware sends structured telemetry frames over UART so the viewer can validate packet boundaries and reject corrupted data. A representative frame contains start-of-frame bytes, payload length, payload fields, and CRC.

Field	Example / Content	Purpose
SOF	0xAA, 0x55	Provides packet alignment and lets the viewer detect frame boundaries.
Length	Payload byte count	Prevents parsing the wrong number of bytes.
Payload	Cell voltages, temperatures, pack current, pack voltage, SOC, SOH, fault flags	Carries BMS state to the viewer.
CRC	Computed over packet data	Detects corrupted or incomplete frames.

The BMS Viewer accepts packets only when SOF, length, and CRC checks pass. This prevents stale or corrupted serial data from being displayed as valid battery information.

Fault Logic

Fault detection is implemented in firmware by comparing validated measurements against configured thresholds. The BMS fault output asserts when a safety violation or communication fault is detected.

Fault = OV OR UV OR OC OR OT OR COMM_LOSS

Fault Type	Trigger Source	Design Response
OV	Cell voltage above overvoltage threshold.	Assert BMS fault and report fault flag in telemetry.
UV	Cell voltage below undervoltage threshold.	Assert BMS fault and report fault flag in telemetry.
OC	Measured pack current above current limit.	Assert BMS fault and report fault flag in telemetry.
OT	Selected thermistor group indicates unsafe temperature.	Assert BMS fault and report fault flag in telemetry.
COMM_LOSS	Invalid or missing LTC6811 communication.	Assert BMS fault and prevent stale data use.

Timing and Verification

Requirement	Measured / Verified Behavior
UART telemetry rate	Stable 10 Hz transmission with valid CRC and no observed packet corruption.
BMS Viewer update rate	Approximately 1.1 Hz displayed update rate.
Packet parsing	>99% parsing accuracy; invalid packets rejected.
isoSPI reliability	500 LTC6811 read cycles completed with 0 CRC errors.
Fault output timing	Fault output asserted low within 20 ms for tested OV, UV, OC, and communication fault cases.
Viewer fault indication	Visual indication occurred within approximately 300-500 ms.

The split between fast embedded fault assertion and slower human-facing display is intentional. The microcontroller fault output provides rapid protection signaling, while the viewer provides diagnostic visibility for testing and maintenance.